

# Main Checklist for Data\_Project

Progress tracking for the DDR Data Logger project on Zybo Z7-20

## Main Checklist

| Stage | Stage Name                            | Status    | Stage Output                                                               |
|-------|---------------------------------------|-----------|----------------------------------------------------------------------------|
| 0     | GitHub + Project Structure            | ✓ Done    | Repository, folders, README, docs, first commit                            |
| 1     | Data Protocol                         | ✓ Done    | docs/data_protocol.md with Header, Length, Data, Checksum                  |
| 2     | Data Generator + Testbench            | ✓ Done    | rtl/data_generator.v, tb/tb_data_generator.v, simulation, waveform, report |
| 3     | Packet Builder FSM + Testbench        | □ Pending | packet_builder_fsm.v + testbench + simulation                              |
| 4     | Sync FIFO + Testbench                 | □ Pending | sync_fifo.v + testbench + simulation                                       |
| 5     | AXI-Stream Master + Testbench         | □ Pending | axis_stream_master.v + testbench + simulation                              |
| 6     | Full RTL Top Simulation               | □ Pending | data_project_top.v + full testbench + simulation                           |
| 7     | Vivado Block Design + AXI DMA + DDR   | □ Pending | Connected Block Design ready to run                                        |
| 8     | Vitis C Verification                  | □ Pending | C software that starts DMA and verifies DDR data                           |
| 9     | Async FIFO + CDC + Testbench          | □ Pending | async_fifo.v + testbench with two clocks                                   |
| 10    | Full Integration with CDC + DMA + DDR | □ Pending | Full system with two clock domains and DDR                                 |
| 11    | ILA + Vivado Reports                  | □ Pending | ILA, timing report, utilization report                                     |
| 12    | Python CSV / Graph                    | □ Pending | CSV, graph, and documentation                                              |
| 13    | UART or SPI Extension                 | □ Pending | Real data source replacing Data Generator                                  |
| 14    | Portfolio Polish                      | □ Pending | GitHub, README, images, explanations, and reports polish                   |

Current status: Stages 0-2 are complete. The next stage is Stage 3 - Packet Builder FSM + Testbench.

# Detailed Checklist for Work Tracking

## Stage 0 - GitHub + Project Structure

✓ Done

- |                                                                        |                                                                |
|------------------------------------------------------------------------|----------------------------------------------------------------|
| <input checked="" type="checkbox"/> Created Data_Project folder        | <input checked="" type="checkbox"/> Created work plan document |
| <input checked="" type="checkbox"/> Created GitHub repository          | <input checked="" type="checkbox"/> Added block diagram        |
| <input checked="" type="checkbox"/> Created README.md                  | <input checked="" type="checkbox"/> Completed first commit     |
| <input checked="" type="checkbox"/> Created organized folder structure | <input checked="" type="checkbox"/> Completed push to GitHub   |

## Stage 1 - Data Protocol

✓ Done

- |                                                                   |                                                                  |
|-------------------------------------------------------------------|------------------------------------------------------------------|
| <input checked="" type="checkbox"/> Created docs/data_protocol.md | <input checked="" type="checkbox"/> Defined Checksum             |
| <input checked="" type="checkbox"/> Defined DATA_WIDTH = 32       | <input checked="" type="checkbox"/> Defined Expected Packet      |
| <input checked="" type="checkbox"/> Defined Header                | <input checked="" type="checkbox"/> Defined PASS / FAIL Criteria |
| <input checked="" type="checkbox"/> Defined Length                | <input checked="" type="checkbox"/> Completed commit             |
| <input checked="" type="checkbox"/> Defined Data sequence         | <input checked="" type="checkbox"/> Completed push               |

## Stage 2 - Data Generator + Testbench

✓ Done

- |                                                                  |                                                                                                                |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> Defined block goal           | <input checked="" type="checkbox"/> Verified expected sequence: 1, 7, 8, 3, 6, 3                               |
| <input checked="" type="checkbox"/> Defined inputs and outputs   | <input checked="" type="checkbox"/> Saved waveform screenshot: images/<br>stage2_data_generator_waveform.png   |
| <input checked="" type="checkbox"/> Wrote rtl/data_generator.v   | <input checked="" type="checkbox"/> Created simulation report: reports/<br>stage2_data_generator_simulation.md |
| <input checked="" type="checkbox"/> Wrote tb/tb_data_generator.v | <input checked="" type="checkbox"/> Completed commit                                                           |
| <input checked="" type="checkbox"/> Ran Behavioral Simulation    | <input checked="" type="checkbox"/> Completed push                                                             |

## Stage 3 - Packet Builder FSM + Testbench

□ Pending

- |                                                           |                                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------|
| <input type="checkbox"/> Define block goal                | <input type="checkbox"/> Run simulation                           |
| <input type="checkbox"/> Define FSM states                | <input type="checkbox"/> Verify Header / Length / Data / Checksum |
| <input type="checkbox"/> Write rtl/packet_builder_fsm.v   | <input type="checkbox"/> Complete commit                          |
| <input type="checkbox"/> Write tb/tb_packet_builder_fsm.v | <input type="checkbox"/> Complete push                            |

## Stage 4 - Sync FIFO + Testbench

□ Pending

- |                                                  |                                                    |
|--------------------------------------------------|----------------------------------------------------|
| <input type="checkbox"/> Define block goal       | <input type="checkbox"/> Verify full/empty         |
| <input type="checkbox"/> Write rtl/sync_fifo.v   | <input type="checkbox"/> Verify overflow/underflow |
| <input type="checkbox"/> Write tb/tb_sync_fifo.v | <input type="checkbox"/> Complete commit           |
| <input type="checkbox"/> Verify write/read       | <input type="checkbox"/> Complete push             |

# Detailed Checklist for Work Tracking

## Stage 5 - AXI-Stream Master + Testbench

☐ Pending

- |                                                                 |                                              |
|-----------------------------------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Define block goal                      | <input type="checkbox"/> Verify backpressure |
| <input type="checkbox"/> Write rtl/axis_stream_master.v         | <input type="checkbox"/> Complete commit     |
| <input type="checkbox"/> Write tb/tb_axis_stream_master.v       | <input type="checkbox"/> Complete push       |
| <input type="checkbox"/> Verify tvalid / tready / tdata / tlast |                                              |

## Stage 6 - Full RTL Top Simulation

☐ Pending

- |                                                         |                                          |
|---------------------------------------------------------|------------------------------------------|
| <input type="checkbox"/> Write rtl/data_project_top.v   | <input type="checkbox"/> Save result     |
| <input type="checkbox"/> Write tb/tb_data_project_top.v | <input type="checkbox"/> Complete commit |
| <input type="checkbox"/> Connect all blocks             | <input type="checkbox"/> Complete push   |
| <input type="checkbox"/> Run full simulation            |                                          |

## Stage 7 - Vivado Block Design + AXI DMA + DDR

☐ Pending

- |                                                          |                                               |
|----------------------------------------------------------|-----------------------------------------------|
| <input type="checkbox"/> Create Block Design             | <input type="checkbox"/> Connect DDR          |
| <input type="checkbox"/> Add Zynq PS                     | <input type="checkbox"/> Pass design validate |
| <input type="checkbox"/> Add AXI DMA                     | <input type="checkbox"/> Complete commit      |
| <input type="checkbox"/> Add Interconnect / SmartConnect | <input type="checkbox"/> Complete push        |
| <input type="checkbox"/> Connect Custom RTL              |                                               |

## Stage 8 - Vitis C Verification

☐ Pending

- |                                                        |                                                                   |
|--------------------------------------------------------|-------------------------------------------------------------------|
| <input type="checkbox"/> Create Vitis project          | <input type="checkbox"/> Read data from DDR                       |
| <input type="checkbox"/> Write DMA initialization code | <input type="checkbox"/> Verify Header / Length / Data / Checksum |
| <input type="checkbox"/> Define DDR buffer address     | <input type="checkbox"/> Complete commit                          |
| <input type="checkbox"/> Start DMA transfer            | <input type="checkbox"/> Complete push                            |

## Stage 9 - Async FIFO + CDC + Testbench

☐ Pending

- |                                                   |                                                       |
|---------------------------------------------------|-------------------------------------------------------|
| <input type="checkbox"/> Write rtl/async_fifo.v   | <input type="checkbox"/> Verify correct data crossing |
| <input type="checkbox"/> Write tb/tb_async_fifo.v | <input type="checkbox"/> Complete commit              |
| <input type="checkbox"/> Verify two clocks        | <input type="checkbox"/> Complete push                |

## Stage 10 - Full Integration with CDC + DMA + DDR

☐ Pending

- |                                                               |                                           |
|---------------------------------------------------------------|-------------------------------------------|
| <input type="checkbox"/> Integrate Async FIFO into the system | <input type="checkbox"/> Verify DDR write |
| <input type="checkbox"/> Update full integration              | <input type="checkbox"/> Complete commit  |
| <input type="checkbox"/> Verify two clock domains             | <input type="checkbox"/> Complete push    |

## Detailed Checklist for Work Tracking

### Stage 11 - ILA + Vivado Reports

☐ Pending

- |                                                  |                                           |
|--------------------------------------------------|-------------------------------------------|
| <input type="checkbox"/> Add ILA                 | <input type="checkbox"/> Save screenshots |
| <input type="checkbox"/> Verify key signals      | <input type="checkbox"/> Complete commit  |
| <input type="checkbox"/> Save timing report      | <input type="checkbox"/> Complete push    |
| <input type="checkbox"/> Save utilization report |                                           |

### Stage 12 - Python CSV / Graph

☐ Pending

- |                                              |                                                   |
|----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Create CSV file     | <input type="checkbox"/> Save screenshot / output |
| <input type="checkbox"/> Write Python script | <input type="checkbox"/> Complete commit          |
| <input type="checkbox"/> Create graph        | <input type="checkbox"/> Complete push            |

### Stage 13 - UART or SPI Extension

☐ Pending

- |                                                    |                                                               |
|----------------------------------------------------|---------------------------------------------------------------|
| <input type="checkbox"/> Choose UART or SPI        | <input type="checkbox"/> Receive real data through the system |
| <input type="checkbox"/> Write communication block | <input type="checkbox"/> Complete commit                      |
| <input type="checkbox"/> Run verification          | <input type="checkbox"/> Complete push                        |

### Stage 14 - Portfolio Polish

☐ Pending

- |                                        |                                                            |
|----------------------------------------|------------------------------------------------------------|
| <input type="checkbox"/> Update README | <input type="checkbox"/> Add explanations for each block   |
| <input type="checkbox"/> Add images    | <input type="checkbox"/> Add Future Work / Lessons Learned |
| <input type="checkbox"/> Add waveforms | <input type="checkbox"/> Prepare project for presentation  |
| <input type="checkbox"/> Add reports   |                                                            |